



AI-Enabled Exoplanet Detection Pipeline

An uncertainty-aware adaptive pipeline designed for AstroSat UVIT photometry. Every detection comes with a calibrated confidence score, full audit trail, and explicit ruling-out of false positives.



FLAGGED FOR HUMAN REVIEW

Candidate: KEPLER-7b (Real Parameters) | True Label: Planet Transit

13.5%

Confidence

1.00

Domain Trust T

0.69

Data Quality

4.88458

d

Period

0.640%

Transit Depth

12.0

BLS SNR

Raw Input Denoising Transit Detection Classification Uncertainty Full Report

Complete Explainability Report

Every decision is traceable. Every alternative is explicitly ruled out. This is what gets sent to the astronomer review queue.

Ruling Out Alternatives

✓ Eclipsing Binary

No secondary eclipse at phase ~0.5. Depth ratio 0.000 (threshold: 0.5)

⚠ Starspot

Starspot probability 49.8%. Transit morphology inconsistent with sinusoidal rotational modulation.

✓ Instrumental Artifact

BLS SNR = 12.0 > 7. Signal present across multiple independent observation windows.

✓ Stellar Blending

No centroid shift detected. Target flux fraction = 0.90. Depth corrected for blending.

Recommended Follow-up

-  Standard follow-up photometry recommended.

Raw Audit Trail (JSON)

```
{
  "candidate_id" : "KEPLER-7b (Real Parameters)"
  "true_label" : "Planet Transit"
  "final_decision" : "FLAGGED FOR HUMAN REVIEW"
  "confidence" : 0.135
  "confidence_interval" : {
    "lower" : 0
    "upper" : 0.3779
  }
  "domain_trust_T" : 0.9991
  "trust_level" : "HIGH"
  "transit_parameters" : {
    "period_days" : 4.88458
    "depth_percent" : 0.6398
    "duration_hours" : 2.88
    "bls_snr" : 11.97
  }
  "classification" : {
    "predicted" : "Starspot"
    "probabilities" : {
      "Planet Transit" : 0.1522
      "Eclipsing Binary" : 0.0748
      "Starspot" : 0.498
      "Instrumental" : 0.275
    }
    "model_agreement" : false
    "sigma_classifier" : 0.1151
  }
  "uncertainties" : {
    "sigma_denoise" : 0.1458
    "sigma_blend" : 0.1111
    "sigma_BLS" : 0.0771
    "sigma_classifier" : 0.1151
    "sigma_total" : 0.1128343110813926
    "dominant_source" : "blend"
  }
  "denoising" : {
    "agreement_score" : 0.8542
    "method" : "ensemble (Savitzky-Golay + Wavelet + Median Filter)"
    "interpretation" : "High confidence – three methods agree"
  }
}
```

